

Bursting in Neurons

Nonlinear Dynamics Final Project

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Introduction

Neuroscience has provided us with several mathematical models to describe the dynamics of neurons. When creating these models, it is important to consider the balance between computational efficiency and biological realism [1]. One well-known example of this balance is demonstrated in the Hodgkin-Huxley model. The model provides an accurate representation of neuronal membrane voltages but its complexity makes it impractical for applications involving numerous neurons [2, 3].

In this paper, we will describe a three-variable ordinary differential equation (ODE) model with a minimal mechanism for bursting, which we will simply refer to as 'the bursting model'. Bursting is a process where neurons fire off signals in quick bursts or groups, rather than one by one. Neural bursts have long been suspected of forming the basic building blocks of neuronal information [4, 5]. Recent research also hints at bursting neurons' capacity to approximate backpropagation and provide a physiological basis for learning [6].

The bursting model we will look at has been applied in the neuromechanical modelling of central pattern generators and bursting motoneurons [7], allowing for a better understanding of the dynamics of legged locomotion [8]. The bursting model also proves interesting within computational neuroscience. It has been proposed as a suitable candidate model for characterizing the general parameter space structure of [9]. Exploring this could improve our understanding of the general theory of neural circuits.

1 Analytical Analysis

1.1 Defining the bursting model

The neural dynamics are defined by the following equations:

$$C\dot{V} = I - g_L(V - E_L) - g_{Na}m_\infty(V)(V - ENa) - g_Kn(V - E_K) - g_{slow}n_s(V - E_K) \quad (1)$$

$$\dot{n} = (n_\infty(V) - n) \frac{1}{\tau_n} \quad (2)$$

$$\dot{n}_s = (n_{s,\infty}(V) - n_s) / \tau_{slow} \quad (3)$$

The dynamics of the gating variables n and n_s are expressed in terms of sigmoid functions of the form $f_\infty = (1 + \exp((V_{half} - V)/k))^{-1}$, with parameters k and V_{half} .

This model uses a rapid(n) and a slow (n_s) recovery variable, allowing for time-scale separation. To take advantage of this, we will take n_s as a constant and analyze the fast subsystem (1) and (2) for different values of n_s . This will help us explain how the system can transition from quiescent to bursting.

Dimensionality Analysis

The dimension of the sigmoid function is given by:

$$f_{\infty} = (1 + \exp(([\text{mV}] - [\text{mV}]) / [\text{mV}]))^{-1} = [1]$$

1. Substituting this and the units associated with the symbols in (1) gives us:

$$[\text{nF}] \frac{[\text{mV}]}{[\text{ms}]} [\text{nA}] - [\mu\text{S}]([\text{mV}] - [\text{mV}]) - [\mu\text{S}][1]([\text{mV}] - [\text{mV}]) - [\mu\text{S}][1]([\text{mV}] - [\text{mV}]) - [\mu\text{S}][1]([\text{mV}] - [\text{mV}])$$

Which gives each term in $[\text{nA}]$. For the equations (2) and (3) we get:

$$\frac{[1]}{[\text{ms}]} = \frac{([1] - [1])}{[\text{ms}]}$$

So all the units are chosen appropriately. We can see that the slow variable n_s acts as an offset current I_{n_s} . We expect this current will act as a bifurcation parameter

Nullclines & Fixed Points

The nullclines of the fast subsystem are given by:

$$V^* = \frac{I + g_{Na}m_{\infty}(V)E_{Na} + g_K n E_K + g_{\text{slow}} n_s E_K + g_L E_L}{g_L + g_{Na}m_{\infty}(V) + g_K n + g_{\text{slow}} n_s} \quad (4)$$

$$n^* = n_{\infty}(V) \quad (5)$$

Fixed points are where the two nullclines intersect each other. Substituting equation (5) in equation (4) gives us:

$$V^* = \frac{I + g_{Na}m_{\infty}(V^*)E_{Na} + g_K n_{\infty}(V^*)E_K + g_{\text{slow}} n_s E_K + g_L E_L}{g_L + g_{Na}m_{\infty}(V^*) + g_K n_{\infty}(V^*) + g_{\text{slow}} n_s} \quad (6)$$

All the V^* that satisfy this equation are fixed points. We can't solve this analytically so we will have to use numerical methods to find the fixed points.

Obtaining the Jacobian

For this part, it is useful to make use of the relation:

$$\frac{\delta f_{\infty}(V)}{\delta V} = \frac{e^{\frac{V_{\text{half}} - V}{k}}}{\frac{1}{k} \cdot \left(e^{\frac{V_{\text{half}} - V}{k}} + 1 \right)^2} = k \cdot f_{\infty}^2(V) \cdot e^{\frac{V_{\text{half}} - V}{k}} = k f_{\infty}(V)(1 - f_{\infty}(V))$$

The Jacobian elements

$$J = \begin{bmatrix} \frac{\partial \dot{V}(V, n)}{\partial V} & \frac{\partial \dot{V}(V, n)}{\partial n} \\ \frac{\partial \dot{n}(V, n)}{\partial V} & \frac{\partial \dot{n}(V, n)}{\partial n} \end{bmatrix} \quad (7)$$

are then given by:

$$\frac{\delta \dot{V}}{\delta V} = \frac{1}{C} [g_L - g_{Na} m_{\infty}(V) (1 + k_m(1 - m_{\infty}(V))(V - E_{Na})) - g_k n - g_s n_s] \quad (8)$$

$$\frac{\delta \dot{V}}{\delta n} = -g_k(V - E_k) \quad (9)$$

$$\frac{\delta \dot{n}}{\delta V} = \frac{1}{\tau_n} k_n n_{\infty}(V) (1 - n_{\infty}(V)) \quad (10)$$

$$\frac{\delta \dot{n}}{\delta n} = -\frac{1}{\tau_n} \quad (11)$$

Stability analysis

We can analyze the stability of the fixed points found from (6) by analyzing their trace and determinant [10]. The trace and determinant of the Jacobian are given by:

$$Tr = \frac{\delta \dot{V}(n_s)}{\delta V} - \frac{1}{\tau_n} \quad (12)$$

$$D = \frac{1}{\tau_n} \left[\frac{\delta \dot{V}(n_s)}{\delta V} + k_n n_{\infty}(V) (1 - n_{\infty}(V)) * g_k (V - E_k) \right] \quad (13)$$

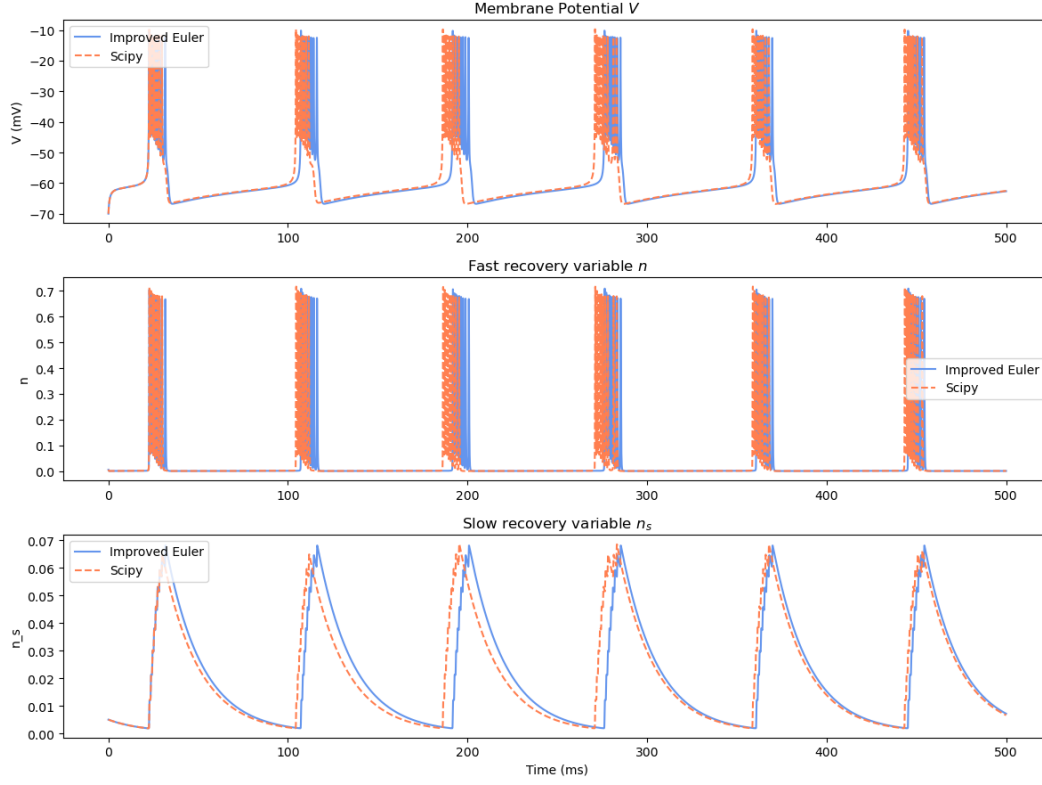
For readability, we kept $\frac{\delta \dot{V}(n_s)}{\delta V}$ in the simplified form. These equations do not simplify any further. Since we cannot analytically derive the fixed points, we will use numerical methods to find the fixed points and then use the eigenvalues of the Jacobian to determine their stability. During the phase plane analysis section, we will explain how the stability of the fixed points changes as we vary the value of n_s and what kind of bifurcations occur.

2 Numerical Analysis

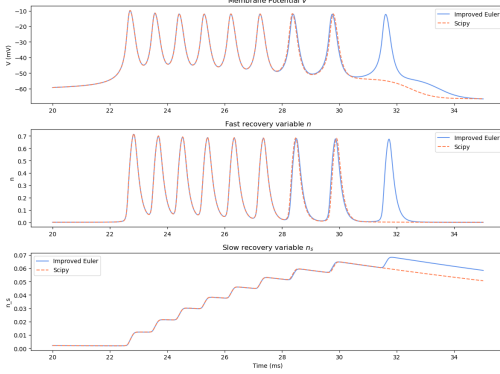
2.1 Methods

The 2 and 3-variable bursting models were implemented in Python. The code for all the figures in this paper can be found on Github [11]. The improved Euler method was implemented as a trial ODE and compared to the SciPy ODE library routine. A brute force algorithm was used to find the optimal time step for the improved Euler method. This brute force method minimizes the absolute error of the variable trajectories compared to the SciPy ODE library routine. The comparison is shown in Figure 1. We will continue using the SciPy ODE to make sure we obtain the most accurate results. The SciPy ODE also has the advantage of being much faster than the improved Euler method.

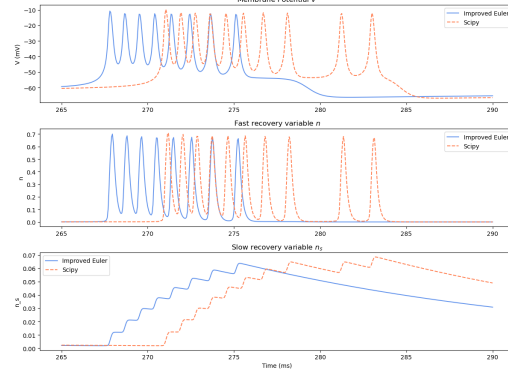
The fixed points were found using a SciPy root-finding algorithm. The Sympy library was used to determine the stability of the fixed points using an analysis of the Jacobians eigenvalues. The average firing rate was found using a simple algorithm to detect the spikes. Once a threshold of $V = -30\text{mV}$ was passed, it was counted as a spike. Bursts were defined as a sequence of spikes with an inter-spike interval of less than 5ms. The firing rate was determined as the number of spikes divided by the total time. The burst rate was determined as the number of bursts divided by the total time. Further details on the implementation can be found in the Jupyter Notebook [11].



(a) Long-term bursting behaviour



(b) Zoomed in plot of the first burst



(c) Zoomed in plot of the fourth burst

Figure 1: Trajectories of the 3-variable bursting model using the SciPy ODE library routine and the improved Euler method. Initial conditions of $V = -70\text{mV}$, $n = 0.005$, $n_s = 0.005$ were used. (a) The improved Euler method is less accurate than the SciPy ODE. This is because the improved Euler method is only 2nd order accurate for linear ODEs. The bursting model is a nonlinear ODE. (b) The first burst is modelled accurately, only missing the last spike. (c) The improved Euler method is not accurate for long-term behaviour.

2.2 Results

Bifurcation values of the slow variable

To find the values of n_s for which the system transitions, we initialized the V and n in a regime appropriate for the behaviour (quiescent or spiking) and then slowly altered n_s until threshold behaviour occurred (either spiking to quiescent or vice versa). The bifurcation from quiescent to spiking is initialized when n_s becomes smaller than 0.00350. This value was found for initial conditions $V = -65\text{mV}$ and $n = 0.005$. The bursting stops when n_s is increased above 0.0645. This value was found for initial conditions $V = -30\text{mV}$ and $n = 0.6$.

Phase plane analysis

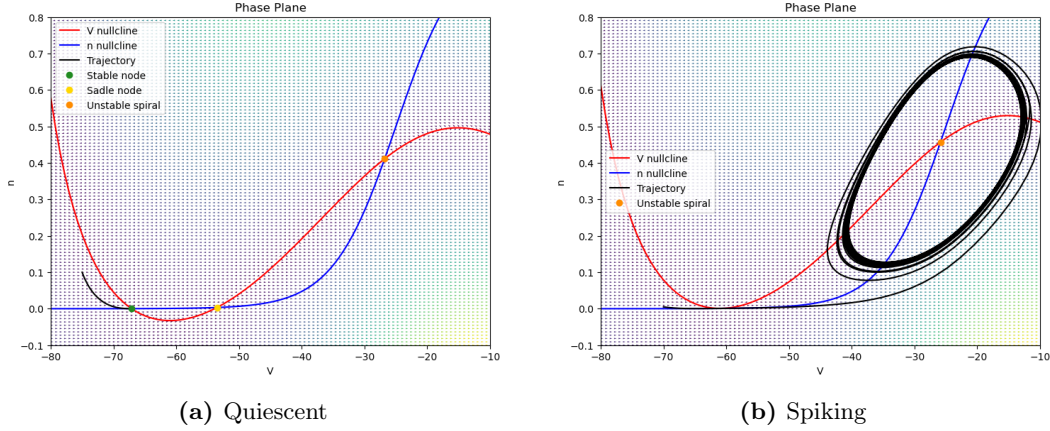


Figure 2: Transition from quiescent to spiking in the phase plane. (a) Trajectory starting at $(-70, 0.05)$ converges to the stable fixed point at $(-67.29986, 0.00022)$ for $n_s = 0.0645$. (b) For critical value $n_s = 0.00350$ the stable fixed point has merged with the saddle point and disappeared leaving a limit cycle around the unstable spiral located at $(-26.8, 0.41)$.

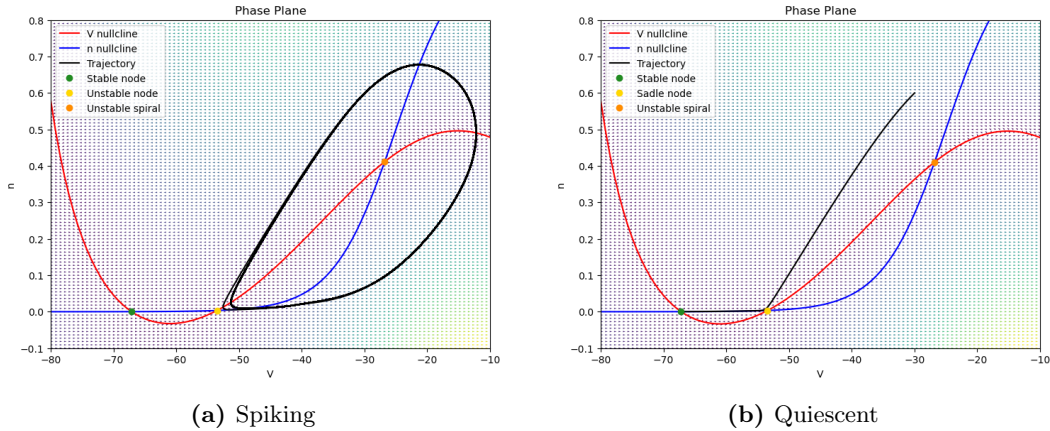


Figure 3: Transition from spiking to quiescent in the phase plane. (a) Trajectory starting in $(-30, 0.6)$ is attracted to saddle node located at $(-53.5, 0.0033)$ but remains in the limit cycle around the unstable spiral $(-26.8, 0.41)$ for a value of $n_s = 0.0644$. (b) Same as in (a) but now the limit cycle hits saddle node and converges to stable node at $(-67.2, 0.000217)$ for a value of $n_s = 0.0645$.

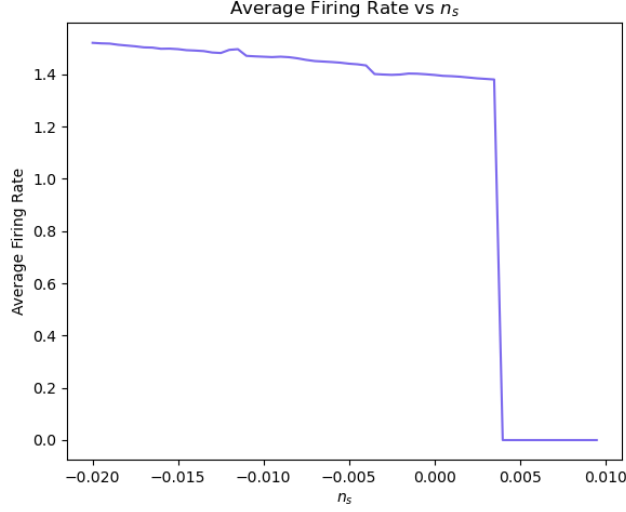


Figure 4: Average firing rate of 2-variable model when varying bifurcation parameter n_s . Initial conditions of $V = -65\text{mV}$ and $n = 0.005$ were used.

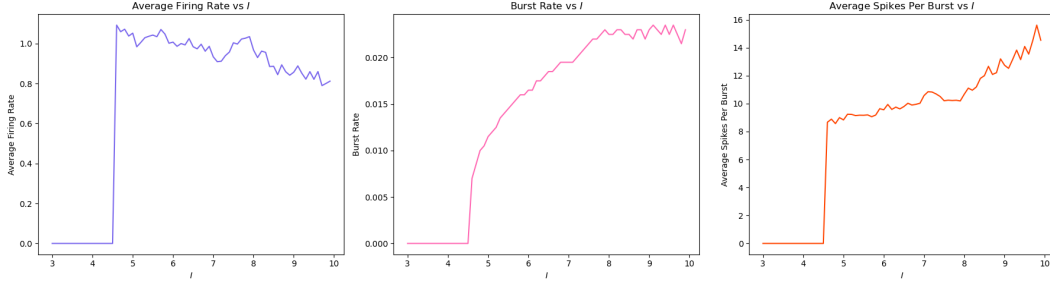


Figure 5: Various rates for the 3-variable model when varying bifurcation parameter I . For all plots initial conditions of $V = -65\text{mV}$, $n = 0.005$ and $n_s = 0.005$ were used. (a) Average firing rate versus I . (b) Average burst rate versus I . (c) Average number of spikes per burst versus I .

In Figure 2 we see that a stable and unstable node merge at the critical value $n_s = 0.00350$. We can classify the bifurcation from quiescent as an SN bifurcation. In Figure 3 we see that for the critical value of $n_s = 0.0645$, the limit cycle hits the saddle node and converges to the stable fixed point. We thus can classify the bifurcation from spiking to quiescent as a homoclinic bifurcation.

The overall behaviour of the 3-variable bursting model is as follows: for $n_s > 0.00350$ the system is in a quiescent state and converges to the stable node. At $n_s = 0.00350$ the stable and unstable node merge and disappear. The system now enters a limit cycle around the unstable spiral and starts spiking. Because of the spikes, the value of n_s starts increasing again and the stable and unstable node reappears. But now the system is attracted to the saddle node instead of the stable node and thus remains in the limit cycle. This continues until $n_s = 0.0645$ where the limit cycle hits the saddle node and converges to the stable node. The system is now in a quiescent state again and the cycle repeats.

Firing and burst rates

In Figure 4 it is demonstrated that the system can not initiate firing above $n_s = 0.0035$. When decreasing n_s below 0.0035 the firing rate increases linearly. The cut-off in firing is

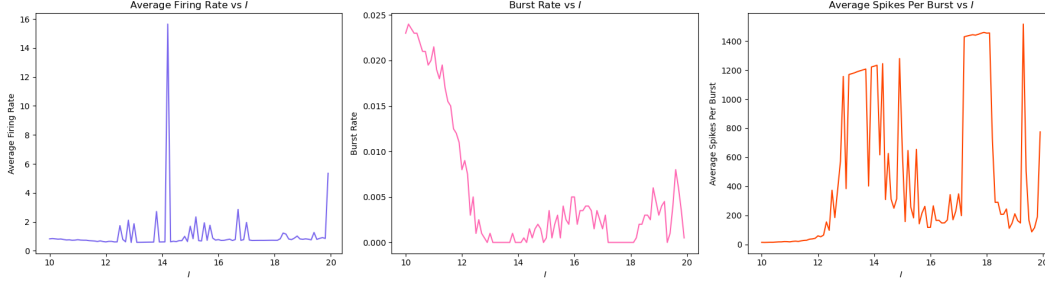


Figure 6: Chaos in the 2-variable model. The firing and bursting rates appear to be random.

step-like, indicating Type II excitability for this model.

In Figure 5 we see that I can also act as a bifurcation parameter. Increasing I gives similar effects as decreasing n_s since both act as an offset to the V -nullcline. At a threshold value of $I = 4.5$ the firing rate sharply increases and then further decreases in a noisy manner. The burst rate appears to increase in a non-trivial way, as does the average spikes per burst. This might be due to the non-linear properties of the system.

Chaos

When looking at the firing and bursting rates for large values of I , we get highly irregular patterns. This is demonstrated in Figure 6. There appears to be no regular pattern in the firing and bursting rates. This might be due to chaos in the system but it can also be due to the way the rates are calculated. Further hints of chaos were not found in the model. This might be a topic for future research.

3 Conclusion & Discussion

We have analyzed the dynamics of a 3-variable bursting model. We have shown that this model accurately models bursting behaviour. Furthermore, we have explained how this bursting happens by analyzing the fast subsystem for different values of the slow variable. We have shown that the slow variable acts as a bifurcation parameter. We have also shown that the system undergoes an SN bifurcation from quiescent to spiking and a homoclinic bifurcation from spiking to quiescent. We have also shown that the system has Type II excitability. Finally, we have shown that the system might exhibit chaotic behaviour for large values of I .

These results show that the model is a suitable candidate for a 'minimal burst' model. Such models can be useful for modelling bursts with a minimal number of variables. This can be useful for modelling large networks of neurons or other applications as described in the introduction. Future work might be to investigate how accurately this system models burst compared to biological neuron bursts.

Throughout this project, I have learned to apply the theory of this course to an actual neuron model. I learned that it might be better to first perform numerical analysis for complicated systems. I initially tried to reason from the analytical side but got it completely wrong, misinterpreting a Hopf bifurcation as the cause of spiking/quiescent transitions. Additionally, I have learned how to simulate a neuron model within Python and to make the tools to analyze it properly. Finally, I learned more details about how neuron models work and what all their parameters mean and how they interact with each other.

References

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